

Technical Supplement

Joint Estimation via GMM

The Identification Problem: Circularity

Recall the structural equations:

$$w_t = \psi L_{St} + \varepsilon_t \quad (\text{Supply})$$

$$L_{it} = \phi^d w_t + \lambda_i \eta_t + u_{it} \quad (\text{Demand})$$

- **The Challenge with PCA:** We cannot perfectly extract λ_i via PCA on L_{it} without first knowing the wage effect $\phi^d w_t$.
- **The Circularity:** We cannot know ϕ^d without a valid instrument z_t . But constructing z_t requires knowing λ_i .
- **The Solution:** We must estimate the structural elasticities (ψ, ϕ^d) and the latent factor loadings (λ_i) **jointly**.

The Core Moment Conditions

GMM identifies parameters by matching theoretical moments to sample moments. The core idea is that a valid instrument $z_t(\lambda)$ must be orthogonal to aggregate shocks.

1. Supply Moment (Identifying ψ):

The instrument must be uncorrelated with the supply shock ε_t .

$$\mathbb{E}[(w_t - \psi L_{St})z_t(\lambda)] = 0$$

2. Demand Moment (Identifying ϕ^d):

The instrument must be uncorrelated with the demand shock η_t . Using the equal-weighted average demand:

$$\mathbb{E}[(L_{Et} - \phi^d w_t)z_t(\lambda)] = 0$$

The GMM Objective Function

Let $\theta = (\psi, \phi^d)$ be the structural parameters, and let the sample moments be:

$$g_{1,T}(\theta, \lambda) = \frac{1}{T} \sum (w_t - \psi L_{St}) z_t(\lambda)$$
$$g_{2,T}(\theta, \lambda) = \frac{1}{T} \sum (L_{Et} - \phi^d w_t) z_t(\lambda)$$

Let $g_T(\theta, \lambda) = [g_{1,T}, g_{2,T}]'$. The GMM estimator seeks (θ, λ) to minimize:

$$J(\theta, \lambda) = g_T(\theta, \lambda)' W g_T(\theta, \lambda)$$

where W is a positive definite weighting matrix.

Concentrated GMM: The Iterative Loop (Part 1)

Simultaneous optimization over elasticities and N loadings is difficult. Gabaix and Koijen (2022) use **Concentrated GMM** (Iterative IV).

1. **Initialize:** Start with an initial guess for the elasticities, $\theta^{(0)} = (\psi^{(0)}, \phi^{d(0)})$ (e.g., from OLS).
2. **Profile out Endogeneity:** Given $\phi^{d(k)}$, compute partialled-out labor demand:

$$\tilde{L}_{it}^{(k)} = L_{it} - \phi^{d(k)} w_t = \lambda_i \eta_t + u_{it}$$

3. **Extract Loadings via PCA:** The matrix $\tilde{L}^{(k)}$ now has a pure factor structure. Extract principal eigenvector to get updated loadings $\lambda^{(k+1)}$.

Concentrated GMM: The Iterative Loop (Part 2)

4. **Construct Instrument:** Using $\lambda^{(k+1)}$, compute the projection matrix Q , the optimal weights $\Gamma^{(k+1)} = S'Q$, and the instrument:

$$z_t^{(k+1)} = \sum \Gamma_i^{(k+1)} L_{it}$$

5. **Update Parameters (IV):** Update elasticities by setting sample moments to zero (equivalent to 2SLS using the new instrument):

$$\psi^{(k+1)} = \frac{\sum w_t z_t^{(k+1)}}{\sum L_{St} z_t^{(k+1)}}, \quad \phi^{d(k+1)} = \frac{\sum L_{Et} z_t^{(k+1)}}{\sum w_t z_t^{(k+1)}}$$

6. **Iterate:** Repeat steps 2–5 until parameters converge.

Why GMM?

- **PCA alone is insufficient:** If wages affect labor demand, running PCA directly on raw outcomes mixes the endogenous response with the exogenous macro shock.
- **The GMM Solution:** It acts as an iterative filter.
 - Guesses the wage effect and subtracts it.
 - Runs PCA to find the clean macro shock.
 - Builds the optimal instrument.
 - Checks if the instrument gives the same wage effect back.
- It loops until the system perfectly balances.